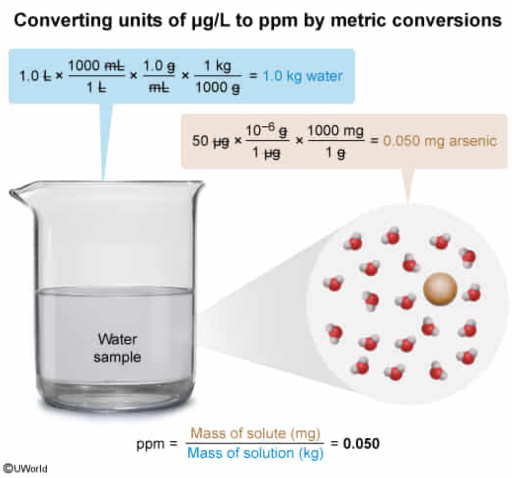
Extremely low concentrations are sometimes expressed in parts per million (ppm), which is equivalent to milligrams of solute per kilogram of solution (mg/kg). If a well water sample contains an arsenic contaminant at a concentration of 50 µg/L, what is the concentration in ppm? (Note: Assume the density of water is 1.0 g/mL.)

- A. 0.000050 [18%]
- B. 0.050 [66%]
- C. 5.0 [6%]
- D. 50 [8%]

Correct
65% answered correctly

Explanation:



When relating the concentration of a solution to a reaction, the concentration is usually best expressed as the moles of solute (ie, a dissolved compound) per liter of solution (ie, a molar concentration). However, in some biological and environmental contexts, it is sometimes convenient to express solution concentrations in terms of the mass of solute per unit volume of solution (eg, g/L, mg/mL).

For very dilute solutions with extremely low concentrations, the value can be alternatively expressed as a mass ratio relative to some basis quantity. For example, comparing the mass ratio using a one-million-unit basis (ie, 1,000,000 = 10^6) yields **parts per million (ppm)**, which is defined as:

$$\text{ppm} = \frac{\text{Mass of solute(g)}}{\text{Mass of solution(g)}} \times 10^6 = \frac{\text{Mass of solute(mg)}}{\text{Mass of solution(kg)}}$$

Given that ppm is equivalent to milligrams of solute per kilogram of solution (mg/kg), the arsenic contaminant concentration of 50.0 µg/L can be easily converted to ppm by applying **metric unit conversions**. Accordingly, the units of the solute mass are converted to grams as:

$$50\,\frac{\cancel{\mu\text{g arsenic}}}{\text{L water}} \times \frac{1 \times 10^{-6}\ \text{g arsenic}}{1\ \cancel{\mu\text{g arsenic}}} = 50 \times 10^{-6}\,\frac{\text{g arsenic}}{\text{L water}} = 0.000050\,\frac{\text{g arsenic}}{\text{L water}}$$

If the units in both the numerator (grams) and the denominator (liters) are scaled by the same factor (1,000) to give units of mg/mL, the factors of 1,000 cancel and yield a result that is numerically the same (ie, g/L = mg/mL).

$$0.000050\,\frac{\cancel{\text{g arsenic}}}{\cancel{\text{L water}}} \times \frac{\cancel{1{,}000}\ \text{mg arsenic}}{1\ \cancel{\text{g arsenic}}} \times \frac{1\ \cancel{\text{L water}}}{\cancel{1{,}000}\ \text{mL water}} = 0.000050\,\frac{\text{mg arsenic}}{\text{mL water}}$$

Because the well water sample is very dilute (ie, almost entirely water), the density of the solution is essentially the same as the density of water. As such, applying this **density as a conversion factor** expresses the concentration as a mass ratio:

$$0.000050\,\frac{\text{mg arsenic}}{\cancel{\text{mL water}}} \times \frac{\cancel{\text{mL water}}}{1.0\ \text{g water}} = 0.000050\,\frac{\text{mg arsenic}}{\text{g water}}$$

Finally, converting the mass of the aqueous solution to kilograms gives the arsenic concentration in ppm (mg/kg):

$$0.000050\,\frac{\text{mg arsenic}}{\cancel{\text{g water}}} \times \frac{1{,}000\ \cancel{\text{g water}}}{1\ \text{kg water}} = 0.050\,\frac{\text{mg arsenic}}{\text{kg water}} = 0.050\ \text{ppm}$$

(Choice A) A value of 0.000050 is equivalent to the concentration expressed in units of milligrams per *gram* (not kilogram). As such, this is not the concentration in parts per million.

(Choice C) A value of 5.0 results from math errors during the calculation.

(Choice D) A value of 50 is the concentration expressed in parts per billion (ie, parts per 10^9 instead of parts per 10^6).

Educational objective:
Measurements of concentration defined based on metric units can be interconverted using metric unit conversions. The density of a substance can be applied as a conversion factor in a calculation.

Time spent:0 sec
Subject:
General Chemistry

QID:403956
System:
Atomic Theory and Chemical Composition